

1 Inverse functions

An inverse of a function f is another function f^{-1} that does the opposite operations in reverse order

- When can inverses exist?

- $f^{-1}(f(x)) = f(f^{-1}(x)) = x$
- How do you find inverse? **ALWAYS** follow *this* process if you want full marks:

1. "Let $y = f^{-1}(x)$ "
2. "Then $x = f(y)$ "
3. (solve for y)
4. "Therefore $y = f^{-1}(x) = \dots$ "

(i) Let $f : D \rightarrow \mathbb{R}, f(x) = x^2 + 6x - 3$

- (a) Find the maximal domain D such that f has an inverse
- (b) Find the inverse f^{-1}
- (c) Find the slope of the tangent to f^{-1} at $x = 10$

(ii) The function defined by $f : A \rightarrow \mathbb{R}$, where $f(x) = -2(x+3)^2 + 4$ will have an inverse function if its domain is

- A. $\mathbb{R}$
- B. $\mathbb{R}^- \cup \{0\}$
- C. $\{x : x < -4\}$
- D. $\{x : x \geq -2\}$
- E. $\{x : x \leq 3\}$

- POI between f and f^{-1} ? Can lie on

$$y = x$$

$$y = -x + c, \quad \text{for some } c$$

- What do the POI on each of those look like? Either (a, a) [on $y = x$] or in pairs of $(a, b), (b, a)$ [on $y = -x + c$]
- $f(x) = x$ gives solutions to all $f(x) = f^{-1}(x)$ ONLY IF f is strictly increasing

1. (see 2003 VCE Mathematical Methods Exam 2 Part 2 Question 4)

a. Sketch the graph of the equation $x = \frac{3t}{5+t^2}$, $t \geq 0$.

b. i. Find, correct to 4 decimal places, the least value of a such that the function

$$h : [a, \infty) \rightarrow \mathbb{R}, \quad h(t) = \frac{3t}{5+t^2}$$

has an inverse function.

ii. For this value of a , sketch the graph of h^{-1} . Label any endpoint with its coordinates. Label any asymptote with its equation.

c. Find the rule for h^{-1} .

2. Consider the function $f : [2, \infty) \rightarrow \mathbb{R}$, $f(x) = \sqrt{x^2 - p}$.

a. Find the value of p if f has an inverse function $f^{-1} : [1, \infty) \rightarrow \mathbb{R}$.

b. Find the rule for f^{-1} .

3. Determine the value(s) of k such that the inverse function of $f : [-k, \infty) \rightarrow \mathbb{R}$, $f(x) = kx^2 + x + 1$ is given by the rule $f^{-1}(x) = \sqrt{2x - 1} - 1$.

4. Let $f : [a, 2) \rightarrow \mathbb{R}$, $f(x) = x^3 - 2x^2 - 4x + 5$ where a is the smallest value for which the inverse function f^{-1} exists. Find the exact value of a .